Algebra 5 — Multiplying brackets · Factorising quadratics · Quadratic equations

No calculator. Marks in brackets. ★ = harder question. Check factorisations by expanding, and solutions by substituting.

1 Expand and simplify: [4]

a $(x+3)(x-5)$ **b** $(x-6)^2$ **c** $(x-1)(x+2)(x-3)$

2 Factorise: [5]

a x^2-4x **b** x^2-49 **c** $x^2-7x+10$

3 Factorise and solve for x: [8]

a $x^2+5x+6=0$ **b** $x^2-3x=10$ **c** $x^2+4x=0$ **d** $x^2-9=0$

4 A triangle has base $2x$ cm and height $(x+3)$ cm. [4]

a Find and simplify an expression for its area.

b The area is 40 cm². Find x.

5 A rectangle $(2x+1)$ cm by $(x+2)$ cm has a square of side x cm cut out of it. [4]

a Find and simplify an expression for the remaining (shaded) area.

b The shaded area is 16 cm². Find the dimensions of the rectangle.

6 Two consecutive positive integers have a product of 132. Form a quadratic equation and find the integers. [3]

Total: 28 marks

Check: FOIL — First, Outside, Inside, Last · $(x-a)^2$ has a middle term · to solve, get "$=0$" first, factorise, then set each bracket to zero · reject solutions that don't fit the context (negative lengths).